Manuscript.docx

by Pakar Online

Type of Manuscript:
Research

Susceptibility Status of Aedes spp. to Pyrethroid Insecticides in Two Dengue-Endemic Subdistricts of Kendari, Indonesia: A Comparative Biochemical Study

Mubarak[1*], Intan Rantetandung[1], Arimaswati[1], Abdul Rahim Sya'ban[2]

ABSTRACT

Introduction: Dengue remains a major public health problem in Indonesia, including Kendari City, with increasing incidence reported in recent years. Vector control through insecticide-based fogging is widely implemented, with pyrethroids as the primary insecticide used. However, continuous exposure may alter mosquito susceptibility and reduce control effectiveness. This study aimed to assess and compare the susceptibility status of *Aedes spp.* mosquitoes in Poasia and Baruga Subdistricts, Kendari.

Methods: An analytical observational study with a cross-sectional design was conducted. Mosquito samples were collected from selected villages in both subdistricts. Susceptibility status was determined based on non-specific esterase enzyme activity using a biochemical assay measured by an ELISA reader. Statistical analysis was performed using Mann–Whitney and Kruskal–Wallis tests.

Results: All mosquito populations from both subdistricts were classified as sensitive to pyrethroid insecticides, with mean absorbance values below 0.700. A significant difference in non-specific esterase enzyme activity was observed between Poasia and Baruga Subdistricts ($p < 0.001$). Significant differences were also found among villages within both subdistricts ($p < 0.001$).

Conclusions: *Aedes spp*. populations remain susceptible to pyrethroid insecticides. However, variations in enzyme activity indicate the need for continuous resistance surveillance to ensure the effectiveness of dengue vector control programs.

Keywords: Aedes *spp*.; dengue; insecticide susceptibility; pyrethroids; vector control

INTRODUCTION

Dengue is one of the most rapidly spreading mosquito-borne viral diseases and remains a major global public health concern, particularly in tropical and subtropical regions. The World Health Organization (WHO) has reported a substantial increase in dengue incidence worldwide over the past decades, with Southeast Asia, including Indonesia, contributing significantly to the global burden.[1,2] In Indonesia, dengue continues to pose a serious health challenge, with recurrent outbreaks and increasing case numbers reported annually.[3]

Kendari City, located in Southeast Sulawesi, is among the dengue-endemic areas experiencing a rising trend in incidence, particularly in Poasia and Baruga Subdistricts.[3] The persistence of dengue transmission in these areas highlights the importance of effective vector control strategies to reduce disease burden and prevent outbreaks.

Vector control remains the primary strategy for dengue prevention, as no universally effective antiviral treatment is available and vaccine coverage is still limited in many endemic regions. Among various control methods, insecticide-based fogging is widely implemented to reduce adult mosquito populations. Pyrethroids are the most commonly used class of insecticides due to their high efficacy, rapid knockdown effect, and relatively low toxicity to humans.[2] However, the extensive and repeated use of pyrethroids may exert selective pressure on mosquito populations, leading to alterations in susceptibility and the eventual development of insecticide resistance.[4]

Insecticide resistance in *Aedes* spp. is a growing concern, as it can significantly compromise the effectiveness of vector control programs and contribute to increased dengue transmission. One of the key mechanisms underlying resistance is metabolic detoxification, which involves elevated activity of enzymes such as esterases, monooxygenases, and glutathione S-transferases.[5,8] Among these, non-specific esterase activity has been widely used as a biochemical marker for early detection of resistance development in mosquito populations.

Monitoring the susceptibility status of mosquito vectors is therefore essential to ensure the continued effectiveness of insecticide-based interventions and to support evidence-based decision-making in vector control programs. Early detection of biochemical changes can serve as an important warning signal before phenotypic resistance becomes evident, allowing timely implementation of resistance management strategies.

Despite the importance of resistance surveillance, data on insecticide susceptibility of *Aedes* spp. in Kendari City remain limited, particularly those based on biochemical approaches. Therefore, this study aimed to assess and compare the susceptibility status of *Aedes* spp. mosquitoes in Poasia and Baruga Subdistricts using biochemical testing based on non-specific esterase enzyme activity.

METHODS

Study Design

This study employed an analytical observational approach using a cross-sectional design to evaluate the susceptibility status of *Aedes* spp. mosquitoes in selected dengue-endemic areas.

Study Area and Period

The research was conducted in two dengue-endemic subdistricts, namely Poasia and Baruga, located in Kendari City, Southeast Sulawesi, Indonesia. Data collection and laboratory analysis were carried out throughout 2024.

Ethical Considerations

The study protocol received ethical clearance from the Institutional Ethics Committee (No. 126/UN29.17.1.3/ETIK/2024). All procedures were performed in accordance with established ethical standards.

Sample Selection

Adult *Aedes* spp. mosquitoes were collected from several selected villages within each subdistrict. Only intact specimens were included for analysis, while damaged or incomplete samples were excluded to ensure data reliability.

Sampling Technique and Sample Size

Mosquito collection was performed using a purposive sampling method. A total of nine villages were included in the study, and from each village, 30 adult mosquitoes were obtained, resulting in a representative sample for comparative analysis.

Biochemical Assay Procedure

Each mosquito specimen was processed individually. The samples were homogenized in phosphate-buffered saline to obtain a uniform suspension. Subsequently, the homogenate was reacted with α-naphthyl acetate as a substrate, followed by the addition of a coupling reagent to facilitate color development. The enzymatic activity was then quantified by measuring absorbance using an ELISA reader at a wavelength of 450 nm.

Based on absorbance values, susceptibility status was categorized into three groups: sensitive ($AV < 0.700$), tolerant (0.700–0.900), and resistant ($AV > 0.900$).

Statistical Analysis

Data analysis was conducted using non-parametric statistical methods. Differences in enzyme activity between the two subdistricts were examined using the Mann–Whitney test, while variations among villages were analyzed using the Kruskal–Wallis test. A p-value of less than 0.05 was considered statistically significant.

RESULTS

All mosquito populations from Poasia Subdistrict showed mean absorbance values below 0.700, indicating susceptibility to pyrethroid insecticides (Table 1).

Table 1. Biochemical Testing of *Aedes spp.* Mosquitoes from Poasia Subdistrict

Subdistrict	**n**	Mean AV
Andounohu	30	0.330
Rahandouna	30	0.419
Anggoeya	30	0.319
Matabubu	30	0.423
Wundumbatu	30	0.432

Similarly, mosquito populations from Baruga Subdistrict were also classified as sensitive, with all mean absorbance values below 0.700 (Table 2).

Table 2. Biochemical Testing of *Aedes spp.* Mosquitoes from Baruga Subdistrict

Village	n	**Mean** AV
Baruga	30	0.224
Lepo-lepo	30	0.260
Watubangga	30	0.291
Wundudopi	30	0.188

A statistically significant difference in non-specific esterase enzyme activity was observed between Poasia and Baruga Subdistricts ($p < 0.001$). In addition, significant differences in enzyme activity were found among villages within both subdistricts ($p < 0.001$).

DISCUSSION

The present study demonstrated that *Aedes* spp. mosquito populations from both Poasia and Baruga Subdistricts remain susceptible to pyrethroid insecticides, as indicated by absorbance values below the established threshold. These findings suggest that pyrethroid-based fogging interventions are still effective in reducing adult mosquito populations in the study areas.

Despite this overall susceptibility, a statistically significant difference in non-specific esterase enzyme activity was observed between subdistricts and among villages. Although all mosquito populations were still categorized as sensitive, the variation in enzyme activity may reflect early biochemical adaptation within the mosquito populations. Such changes are important because metabolic detoxification, particularly through elevated esterase activity, has been widely recognized as one of the primary mechanisms underlying insecticide resistance in *Aedes* mosquitoes.[5,8]

The absence of resistance in the present study may be associated with the relatively limited duration or intensity of pyrethroid use in Kendari City. Previous studies have reported that resistance in *Aedes aegypti* typically develops following prolonged and repeated exposure to insecticides, often over several years.[11,12] In this context, the continued susceptibility observed in the study area may indicate that the selection pressure has not yet reached a critical level sufficient to induce phenotypic resistance.

In addition, the observed differences in enzyme activity among villages may be influenced by local environmental and operational factors. Variations in insecticide application practices, frequency of fogging, and community-level exposure could contribute to heterogeneous selection pressure across locations. Furthermore, ecological conditions and genetic variability within mosquito populations may also play a role in shaping enzyme activity profiles.[13,14] The movement of mosquitoes between adjacent areas may further facilitate the spread of resistance-associated traits, potentially leading to future resistance development if not properly managed.[15]

From a public health perspective, the findings of this study provide important implications for dengue control programs. The current effectiveness of pyrethroid insecticides supports their continued use in vector control interventions. However, the presence of variation in enzyme activity highlights the need for ongoing surveillance. Early detection of biochemical changes is crucial, as it allows for timely implementation of resistance management strategies, such as rotation of insecticide classes or integration with non-chemical control methods.

It is also important to consider that failure to detect early resistance may compromise vector control effectiveness and increase the risk of dengue transmission. In endemic settings, reduced efficacy of insecticides may lead to higher mosquito survival rates, ultimately contributing to sustained or increased disease incidence.

This study has several limitations. The assessment of susceptibility was based solely on biochemical analysis without confirmation through standard WHO bioassays. While biochemical methods provide valuable insights into underlying resistance mechanisms, they may not fully reflect phenotypic resistance. Therefore, future studies incorporating both biochemical and bioassay approaches are recommended to obtain a more comprehensive understanding of insecticide resistance status.

CONCLUSION

The findings of this study indicate that Aedes spp. mosquito populations in Poasia and Baruga Subdistricts remain susceptible to pyrethroid insecticides, suggesting that current insecticide-based vector control strategies are still effective in these areas.

However, the observed variation in non-specific esterase enzyme activity among locations may represent early biochemical changes that could precede the development of insecticide resistance. This highlights the importance of continuous and systematic surveillance to detect emerging resistance at an early stage.

Regular monitoring, combined with appropriate insecticide management strategies, is essential to maintain the effectiveness of dengue vector control programs and to reduce the risk of future resistance development.

Conflict of Interest

The authors declare that there is no conflict of interest regarding the publication of this study.

Funding

This research received no external funding.

REFERENCES

1. World Health Organization. Dengue – global situation. 2024. Available from: https://www.who.int/emergencies/disease-outbreak-news/item/2023-DON518
2. Kementerian Kesehatan Republik Indonesia. Laporan tahunan demam berdarah dengue. Jakarta: Kementerian Kesehatan RI; 2022.
3. Dinas Kesehatan Kota Kendari. Laporan kasus demam berdarah dengue dan data penggunaan fogging tahun 2019–2024. Kendari: Dinas Kesehatan Kota Kendari; 2024.
4. Mubarak, Satoto TBT, Umniyati SR. Analisis penggunaan insektisida malation dan temefos terhadap vektor demam berdarah dengue Aedes aegypti di Kota Kendari Sulawesi Tenggara. Medula. 2015;2(2):134–142. DOI: 10.33772/medula.v2i2.2542
5. Anindita R, Kesetyaningsih TW. Deteksi resistensi larva Aedes aegypti dengan uji biokimia berdasarkan aktivitas enzim esterase di Kabupaten Bantul DIY. Mutiara Medika. 2007;7(2):88–94. DOI: https://doi.org/10.18196/mmjkk.v7i2%20(s).1661
6. Soderlund DM. Neurotoxicology of pyrethroid insecticides. Cornell Univ Dept Entomol. 2020;4(3):113–165. DOI: https://doi.org/10.1016/bs.ant.2019.11.002
7. Field LM, Davies TGE, O'Reilly AO, Williamson MS, Wallace BA. Voltage-gated sodium channels as targets for pyrethroid insecticides. Eur Biophys J. 2017;46(7):675–679. DOI: https://doi.org/10.1007/s00249-016-1195-1
8. Black WC, Snell TK, Saavedra-Rodriguez K, Kading RC, Campbell CL. From global to local: New insights into features of pyrethroid detoxification in vector mosquitoes. Insects. 2021;12(4):1–20. DOI: https://doi.org/10.3390/insects12040276
9. Zong W, Su W, Xie Q, Gu Q, Deng X, Ren Y, et al. Expression, characterization, and immobilization of a novel SGNH esterase Est882 and its potential for pyrethroid degradation. Front Microbiol. 2022;13:1069754. DOI: https://doi.org/10.3389/fmicb.2022.1069754
10. Putra ER, Ahmad I, Prasetyo DB, Susanti S, Rahayu R, Hariani N. Detection of insecticide resistance in the larvae of some Aedes aegypti (Diptera: Culicidae) strains from Java, Indonesia to temephos, malathion, and permethrin. Int J Mosq Res. 2016;3(3):23–28.
11. Irawati NB, Nur, Putri NE. Resistensi nyamuk Aedes aegypti terhadap cypermethrin di Kabupaten Klaten, Jawa Tengah. J Kesehat Lingkung Ruwa Jurai. 2021;15(1):1–7. DOI: https://doi.org/10.26630/rj.v15i1.2608
12. Sayono, Syafruddin D, Sumanto D. Distribusi resistensi nyamuk Aedes aegypti terhadap insektisida sipermetrin di Semarang. Available from: http://jurnal.uimus.ac.id
13. Handayani D, Zuhirman, Putra RM. Uji resistensi nyamuk Aedes aegypti terhadap sipermetrin 0,05% di Pelabuhan Sungai Duku dan Bandara Sultan Syarif Kasim II Pekanbaru. J Kesehat Sehati. 2021;1(1):16–21. DOI: https://doi.org/10.52364/sehati.v1i1.3
14. Windyaraini D, Hening. Status resistensi terhadap insektisida organofosfat dan profil isoenzim esterase non-spesifik nyamuk *Aedes spp*. Yogyakarta: Universitas Gadjah Mada; 2015.
15. Susanty E, Lesmana SD, Taufik R. Status resistensi Aedes aegypti terhadap malathion 0,8% di Kelurahan Labuh Baru Timur Kecamatan Payung Sekaki Kota Pekanbaru. J Med Udayana. 2023;12(4):38–42. DOI: https://doi.org/10.31964/jkl.v23i1.1082
16. Couso-Ferrer F, Arouri R, Beroiz B, Perera N, Cervera A, Navarro-Llopis V. Cross-resistance to insecticides in a malathion-resistant strain of Ceratitis capitata (Diptera: Tephritidae). J Econ Entomol. 2011;104(4):1349–1356. DOI: https://doi.org/10.1603/EC11082

Manuscript.docx

ORIGINALITY REPORT

17%	11%	14%	4%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Vyas Primaben, Ashish Kumar Singh. "Silent Threat: Insecticide Resistance and Escalating Dengue Cases in India", EPJ Web of Conferences, 2026 Publication	2%
2	jccpractice.com Internet Source	2%
3	pmc.ncbi.nlm.nih.gov Internet Source	1%
4	publishing.emanresearch.org Internet Source	1%
5	www.mdpi.com Internet Source	1%
6	Submitted to Universiti Putra Malaysia Student Paper	1%
7	mattioli1885journals.com Internet Source	1%
8	pubmed.ncbi.nlm.nih.gov Internet Source	1%
9	tpmap.org Internet Source	1%
10	Budiyanti Mulyaningsih, Sri Anum Sari. "Status of Resistance of Aedes sp. Mosquitoes to Malathion 0.8% at Poltekkes Banjarmasin, Banjarbaru", JURNAL KESEHATAN	1%

LINGKUNGAN: Jurnal dan Aplikasi Teknik Kesehatan Lingkungan, 2026
Publication

11	Fatemeh Mohammadi, Seyedamir Tabibzadeh Dezfooli, Babak Sadeghzadeh, Mojtaba Chardoli et al. "Determining the Extent of Agreement for Determining the Echographic Ejection Fraction Determined by Emergency Medicine Residents Compared with a Cardiologist", Journal of Diagnostic Medical Sonography, 2023 Publication	1 %
12	www.researchgate.net Internet Source	1 %
13	Hemanga Hazarika, Ravi Kumar Rajan, Padmanath Pegu, Punamjyoti Das. "Insecticide resistance in mosquitoes: molecular mechanisms, management, and alternatives", Journal of Pest Science, 2025 Publication	1 %
14	Mukesh Shakya, Abhijit Nandi, Ashutosh Fular, Sachin Kumar et al. "Synergistic property of piperonyl butoxide, diethyl maleate, triphenyl phosphate and verapamil hydrochloride with deltamethrin and ivermectin against Rhipicephalus microplus ticks", Ticks and Tick-borne Diseases, 2022 Publication	1 %
15	B.K. Tyagi. "Mosquitoes of India - Biology, Phylotaxonomy, Host-Parasite Interaction and Resistance Development", Routledge, 2025 Publication	<1 %
16	Javier Tognarelli, Pablo R. Moya, Christian R. González, Ximena Collao-Ferrada. "Global	<1 %

distribution and impact of knockdown resistance mutations in Aedes aegypti on pyrethroid resistance", Parasites & Vectors, 2025
Publication

17	Liena Sofiana, Rokhmayanti Rokhmayanti, Martini Martini, Dewi Ariyani Wulandari. "Insecticide resistance of Aedes aegypti in Indonesia: a systematic review", International Journal of Public Health Science (IJPHS), 2023 Publication	<1 %
18	Maria Cholvi, Riccardo Moretti, Hugo Costa Osório, Gregory L'Ambert et al. "Present and Future of Mosquito-Borne Disease Control in Europe with a Specific Focus on the Mediterranean", Insects, 2026 Publication	<1 %
19	etd.repository.ugm.ac.id Internet Source	<1 %
20	jurnal.ugm.ac.id Internet Source	<1 %
21	www.frontiersin.org Internet Source	<1 %
22	Jonathan Rene Hernandez, Patricia Victoria Pietrantonio. "Pyrethroid resistance in Aedes aegypti: genetic mechanisms worldwide, and recommendations for effective vector control", Parasites & Vectors, 2025 Publication	<1 %
23	www.ncbi.nlm.nih.gov Internet Source	<1 %

Exclude quotes	On	Exclude matches	Off
Exclude bibliography	On		